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Reliability screening for vehicle routing with time windows: a solver-independent alternative to travel-time padding

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Abstract

Routes planned on nominal travel times run late under congestion and random delay. Padding every trip, the usual remedy, adds distance and vehicles even to safe routes. This paper adds a statistical reliability check to any solver for the vehicle routing problem with time windows. The solver produces candidate plans with internal buffers, leaving customer windows unchanged. A plan is admitted only if a 95% Wilson lower bound on the probability that at least 95% of customers are served on time reaches 0.95, and is then tested on fresh scenarios. Capping each buffer at the customer's direct-reachability limit removes a failure that budgeted-robust models share. On synthetic instances with 20–200 customers, screening beat 20% padding by 8.3–21.4 percentage points, though not significantly at 200 customers. In three timestamped protocols, seven of eight primary hypotheses held; the exception was a conservative-speed fallback. Results depend on the assumed delay model.

Keywords: vehicle routing; time windows; stochastic travel times; time-dependent travel times; chance constraints; simulation; reproducibility

1. Introduction

In the vehicle routing problem with time windows (VRPTW), a fleet of vehicles with limited capacity and shift length must serve each customer within an agreed window (Solomon, 1987; Toth and Vigo, 2014). Most planning tools solve it with a single matrix of travel times. On the road, travel times change through the day and from one day to the next, and a delay early in a route pushes every later customer towards the end of its window. A plan that is optimal on nominal times takes no account of this.

The simplest safeguard is to plan with a lower speed than expected. Padding every trip by a fixed percentage is easy to explain and needs no new software. Its cost is that off-peak trips are padded as much

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as peak ones, so the plan uses extra distance and vehicles whether a route is at risk or not. Stochastic and robust formulations of the VRPTW spend that protection more carefully, because they model the uncertainty inside the optimization (Errico et al., 2018; Agra et al., 2013; Gendreau et al., 2016). They need a solver written for their uncertainty model, however, and most operators run a general-purpose solver that they cannot easily replace.

This paper studies a procedure that keeps the operator's solver and adds a statistical test after it. The solver is asked for several candidate plans, each planned against internal deadlines that have been brought forward by a buffer, and the windows offered to customers do not change. Each candidate is simulated on a set of days drawn from a time-dependent model with correlated delays. It is admitted only if a one-sided Wilson lower bound on the probability of a good service day reaches a target, and the best admitted plan under the planner's own objective is returned. The reliability of that plan is then reported from a second, independent set of simulated days that played no part in choosing it. Because the test comes after the solve, the same rule applies to any solver and any objective.

The paper contributes in three ways.

1. It sets out the screening-and-validation procedure and identifies a failure that internal buffers can cause. When a deadline is brought forward far enough, a customer can become unreachable even by a direct trip from the depot, and the solver then returns no plan at all. Capping each customer's buffer at its direct-reachability limit prevents this. The same failure occurs in a budgeted-robust VRPTW model, and the corresponding relief prevents it there as well.
2. It compares screening with uniform padding on matched instances and reports what the extra reliability costs. In a development study on 53 synthetic instances with 20 to 200 customers, the screened procedure was 8.3–21.4 percentage points more reliable than 20% padding on the 46 instances where padding found a plan, depending on size, although the interval at 200 customers included zero. Three protocols, each timestamped in a public repository with frozen code before any of its runs (Ali, 2026a,b,c), then tested the main findings on new instances with two generators, OR-Tools (Perron and Furnon, 2026) and hybrid genetic search (HGS; Vidal, 2022; Wouda et al., 2024). Seven of the eight primary hypotheses met their decision criteria, and the one that did not is reported alongside the others.
3. It sets the procedure against a budgeted-robust optimizer (Bertsimas and Sim, 2004; Agra et al., 2013) and tests a rule for instances on which no plan passes. Robust plans with reachability relief, put through the same screen, were more reliable on instances with up to 100 customers. They also used more vehicles, and when they were limited to the comparator's fleet their advantage was no longer statistically detectable. Returning the best-screened candidate when nothing is admitted raised the average service over all instances in a prespecified test on 53 further instances.

All instances are synthetic and the delay model is assumed, so the evidence is about repeatability in one controlled setting. Every number in the paper can be regenerated from the archived records. The full model, all result tables and the further checks are in the Supporting Information (SI).

Section 2 reviews related work and Section 3 defines the models and the procedure. Sections 4 and 5 give the design and the results, Section 6 discusses practical use and limitations, and Section 7 concludes.

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2. Related work

Time-dependent travel needs some care, because applying a speed multiplier at the moment of departure can break the first-in-first-out (FIFO) property. The integrated speed profile of Ichoua et al. (2003), which is used here, avoids this, and Ghiani and Guerriero (2014) analyse its properties. For uncertain travel and service times, the stochastic VRPTW literature uses chance constraints, recourse models, approximate feasibility tests and data-driven formulations (Laporte et al., 1992; Kenyon and Morton, 2003; Li et al., 2010; Errico et al., 2018; Bomboi et al., 2022; Serrano et al., 2026). Ehmke et al. (2015) build per-customer service levels into a routing method, Taş et al. (2014) model time dependence, stochastic travel and soft windows together, and Bomboi et al. (2022) check route feasibility when travel times are correlated and time-dependent.

The closest work to the buffers used here is on self-imposed time windows, introduced by Jabali et al. (2015), and on window assignment under stochastic travel times with per-customer chance constraints (Vareias et al., 2019). Both place the buffers inside a dedicated optimization and judge reliability one customer at a time. Neither can be attached to an unmodified solver, and neither bounds the probability that a whole service day goes well.

Robust models require the plan to stay feasible for every realization in an uncertainty set. Agra et al. (2013) solve a robust VRPTW with the budgeted set of Bertsimas and Sim (2004), Jaillet et al. (2016) minimize a requirements violation index, and Zhang et al. (2019) develop a riskiness index together with an exact algorithm. Gendreau et al. (1996) and Gendreau et al. (2016) survey stochastic routing more broadly. The experiments below compare the procedure with padding and with one budgeted robust model. Chance-constrained and distributionally robust VRPTW solvers are not included.

The procedure studied here makes its statistical claim after the solve and builds no feasibility approximation into a specialist search. Routes come from a replaceable nominal solver, a family of internal buffers is searched on one set of simulated days, and the chosen plan is checked once on a held-out set. The service event limits the share of late customers over the whole day, which is closer to the plan-level risk measures of Kenyon and Morton (2003) than to per-customer constraints, and the windows offered to customers never change. Monte Carlo approximations of chance constraints need a separate check of the solution they select (Ahmed and Shapiro, 2008), and the held-out set provides it. The Wilson bound is used because normal-approximation intervals behave poorly for probabilities close to one (Wilson, 1927; Brown et al., 2001).

3. Model and procedure

3.1. Planning model

Let $N = \{0, 1, \ldots, n\}$ be the nodes, with depot 0 and customers C, and let V be a homogeneous fleet with capacity Q and maximum shift duration $H^{\max}$. Customer i has demand δ_i, service time s_i and an offered window $[l_i, u_i]$ for the start of service. Planning uses nominal distances d_{ij} and travel times $\bar{t}_{ij}$. A plan assigns customers to routes, fixes the order of visits and sets a depot departure d_v and return r_v for each vehicle v, with service starting at time a_i at customer i. The full formulation is in Section S1, where $x_{ijv} = 1$ if vehicle v drives arc (i, j) and $y_v = 1$ if vehicle v is used. A plan is described by four

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criteria:

$$F_1 = c_f \sum_v \sum_{i,j} d_{ij} x_{ijv}, \quad F_2 = \sum_{i \in C} \max\{0, s^* - (u_i - a_i)\}, \quad F_3 = \sum_v (r_v - d_v), \quad F_4 = \sum_v y_v. \tag{1}$$

These are the distance cost at c_f per kilometre, the slack deficit against a target slack s^*, the total driver duration and the number of vehicles. Each criterion is scaled by its range $[L_k, U_k]$ over four anchor solutions, each of which minimizes one criterion alone, and the candidate objective is

$$Z = \sum_{k=1}^{4} w_k \, \frac{F_k - L_k}{\max\{U_k - L_k, 1\}}. \tag{2}$$

The weights $\mathbf{w} = (0.5693, 0.2643, 0.1055, 0.0609)$ were fixed before any experiment and are treated as an experimental setting. Their origin is given in Section S12, and their effect is examined in Section 5.6. The objective shapes the candidates and ranks the admitted ones. The admission test does not use it, so any other scalar objective could take its place.

3.2. Evaluation model

The day is divided into periods $[T_k, T_{k+1})$ with speeds $\nu_k > 0$. When a vehicle leaves i for j at time τ, the deterministic travel time is the smallest $\Delta \geq 0$ with

$$\int_{\tau}^{\tau+\Delta} \nu(u)\, du = d_{ij}, \tag{3}$$

which preserves FIFO (Ichoua et al., 2003). In scenario ω this time is multiplied by $\exp\{\varepsilon_{ij}(\omega)\}$, where

$$\varepsilon_{ij}(\omega) = \sigma\big(\sqrt{\rho}\,\xi_0(\omega) + \sqrt{1-\rho}\,\xi_{ij}(\omega)\big), \tag{4}$$

ξ_0 is a standard normal factor shared by all arcs in the scenario and ξ_{ij} is specific to the arc. The multiplier has median one, so the deterministic profile gives the median travel time. A fixed set of simulated days drawn in this way is called a scenario *bank*. Each arc draws from its own random stream, keyed by the bank seed and its endpoints, so plans evaluated on the same bank meet the same delays on the arcs they share. In every scenario each vehicle leaves the depot at its planned time.

3.3. Screening and validation

For plan R and scenario ω, let $q(R, \omega)$ be the share of customers whose service starts by u_i. The service event is $J(R, \omega) = \mathbf{1}\{q(R, \omega) \geq \kappa\}$ with $\kappa = 0.95$, so a day with $J = 1$ is a success, and the target is $\Pr[J = 1] \geq \alpha$ with $\alpha = 0.95$. Asking for every customer to be on time would make a poor target

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on large instances, since that probability falls with the number of customers even for good plans. It is reported, but it is not used for admission.

For each buffer β in the grid, the solver plans against the earlier deadline $\max\{l_i, u_i - \beta\}$ and minimizes (2); the anchor ranges stay at their uncontracted values. Each candidate is simulated on a screening bank of N_s scenarios. If it succeeds on a share $\hat{p}$ of them, its one-sided Wilson lower bound at level $1-\eta$ is

$$L(\hat{p}) = \frac{\hat{p} + z^2/(2N_s) - z\sqrt{\hat{p}(1-\hat{p})/N_s + z^2/(4N_s^2)}}{1 + z^2/N_s}, \qquad z = z_{1-\eta}. \tag{5}$$

A candidate is admitted when $L(\hat{p}) \geq \alpha$, and the admitted candidate with the smallest Z is selected. The selected plan is simulated once more on a larger validation bank that has not been used before, and this run only reports its reliability. Lateness is always scored against the offered windows u_i. If no candidate is admitted, a fallback rule fixed in advance supplies a plan, so the procedure returns something on every instance. The results therefore report the validated reliability of admitted plans separately from the reliability of whatever the procedure returned, fallbacks included. The average of the latter over instances is called *pooled service*. The Wilson bound applies to each candidate on its own, and a family-wise version over the whole grid is examined in Section 5.6. The procedure has four steps.

1. For each buffer $\beta \in \{0, 5, \ldots, 60\}$ minutes, solve the planning model with deadlines $\max\{l_i, u_i - \beta\}$ (or (6) when capped).
2. Simulate each candidate on the screening bank and compute $L(\hat{p})$.
3. If some candidate has $L(\hat{p}) \geq \alpha$, select the admitted candidate with the smallest Z; otherwise apply the fallback rule.
4. Simulate the selected plan once on the validation bank and report its service probability.

3.4. Reachability-capped contraction

If a deadline is brought forward too far, a customer can become unreachable even by a direct trip from the depot, and then no plan exists at that buffer. Let $e_i = \max\{l_i, \bar{t}_{0i}\}$ be the earliest time at which service could start after a direct trip. The capped procedure plans against

$$u_i^{\mathrm{cap}}(\beta) = \min\big\{u_i, \max\{l_i, u_i - \beta, e_i\}\big\}, \tag{6}$$

so contraction alone never makes a reachable customer unreachable. The cap does not guarantee that a joint routing exists or that the target will be met. A shortest-path relaxation certifies when a contracted model is infeasible, which separates proven infeasibility from search failure (Section S4.1).

3.5. Comparators

Conservative speed minimizes distance at 33.3 km/h instead of 40 km/h, which pads every trip by 20%. *Screened conservative speed* applies the same screen to candidates planned at that speed. The *original procedure* screens contracted candidates without the cap, and the *capped procedure* uses (6). Candidates

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Table 1
Instances and parameters of the development study. The confirmatory studies use the same sizes with the seeds shifted (Table 2).

Parameter	Value
Customers (vehicles available; instances)	20 (6; 20), 50 (14; 15), 100 (26; 10), 200 (40; 8)
Nominal speed; shift	40 km/h; 600 min from 06:00
Speed divisors in the shift	0.9 (06:00–07:00), 1.2 (07:00–09:00), 1.0 (09:00–16:00)
Capacity; service time; target slack s^*	1000; 10 min; 15 min
Disturbance σ; latent correlation ρ	0.20; 0.50 (sweep $\sigma \in \{0.1, 0.2, 0.3\}$, $\rho \in \{0, 0.5, 0.9\}$)
Service fraction κ; target α; Wilson η	0.95; 0.95; 0.05 (one-sided)
Screening; validation; sweep scenarios	1000; 5000; 1000 per cell
Buffer grid	$\beta \in \{0, 5, \ldots, 60\}$ min
Solver budgets	OR-Tools 5 s per candidate; HGS 10,000 iterations

come from OR-Tools (Perron and Furnon, 2026) or from the HGS of PyVRP (Wouda et al., 2024). HGS solves the same planning model, but it cannot include the slack-deficit term while it generates plans.

The robust comparator is a VRPTW with budgeted uncertainty (Bertsimas and Sim, 2004; Agra et al., 2013). The travel time of each arc may rise from $\bar{t}_{ij}$ to $\bar{t}_{ij} + \lceil \theta \bar{t}_{ij} \rceil$, and a route must remain feasible whenever at most Γ of its arcs take their upper value. Under *box uncertainty* all of them may. Feasibility is checked exactly with the dynamic programme of Agra et al. (2013) inside a large neighbourhood search written for this study. The search minimizes distance and stops after a fixed number of iterations, so its output is deterministic. Robust models run into the same reachability problem as contraction, since a customer that cannot be reached in time by a direct trip at its upper travel time makes every plan infeasible. The *relieved* variant raises that customer's robust deadline to its worst-case direct start, while every original window must still hold under nominal travel times. *Screened robust* passes all robust plans for an instance through the same screen. It returns the shortest admitted plan, or the plan with the highest screening success if none is admitted. Section S8 gives the details.

Three fallback rules are compared. The *slack-aware* rule returns the uncontracted plan of the weighted objective, the *conservative* rule returns the generator's conservative-speed plan, and the *best-screened* rule returns the candidate of the procedure's own family with the highest screening success. None of them looks at a validation result. Unless stated otherwise, the OR-Tools procedures use the slack-aware fallback and the HGS procedures the conservative one, as their protocols specify.

4. Experimental design

Each instance has a central depot and customers spread uniformly over a 100×100 grid. Demands are uniform on 10–49 units, service takes 10 minutes, window widths are uniform on 90–149 minutes, vehicles carry 1000 units and the shift lasts 600 minutes from 06:00. Table 1 lists the instance sizes and the model constants. Speeds drop during a morning peak from 07:00 to 09:00, and the speed profile continues into the evening so that routes running past the end of the shift are still evaluated.

In the development study every configuration was run on 53 seeded instances, and its findings are treated as exploratory. All configurations of an instance share the same screening and validation banks, so every comparison is paired. Intervals across instances come from a paired bootstrap with 20,000 resamples and a fixed seed. A configuration that finds no plan scores zero on that instance, so a failure

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Table 2
The studies. Seeds of the confirmatory instances are the development seeds shifted by a fixed offset per study.

Study	Instances	Generators	Question
Development	53	OR-Tools	Screening versus padding; capping; robust comparator (exploratory)
Protocol 1 (H1–H4)	53 new	OR-Tools	Do screening and capping beat padding on new instances?
Protocol 2 (G1–G4)	53 new	HGS, OR-Tools	Does the gain survive a stronger generator? Does a conservative fallback help?
Protocol 3A (R1–R4)	90 earlier	Robust, HGS, OR-Tools	Are screened robust plans more reliable, and at what fleet cost?
Protocol 3B (F1–F4)	53 new	OR-Tools, HGS	Does the best-screened fallback raise service?

counts against the configuration that failed. Software errors are logged and stop the analysis, and they are never scored as routing outcomes.

Three confirmatory protocols followed (Table 2). Each protocol fixed its hypotheses, tests, decision rules, failure policy and instance seeds. It was committed to a public repository, together with a SHA-256 manifest of the code that runs and analyses it, and archived on Zenodo (Ali, 2026a,b,c), all before any of its instances was solved. The analysis programs refuse to run unless every record is present and every frozen file matches its manifest. Protocol 1 tested the OR-Tools procedures on 53 new instances. Protocol 2 used 53 further instances to test HGS as the generator and the conservative fallback. Protocol 3 had two parts: part A repeated the robust comparison on the 90 earlier confirmatory instances with up to 100 customers, and part B tested the best-screened fallback on 53 new instances. Within each study the two primary hypotheses use 97.5% intervals (a Bonferroni adjustment), and each is confirmed only if its interval excludes zero in the stated direction. Secondary hypotheses use 95% intervals and cannot rescue a primary one. Since every instance comes from the same generator and delay model, the protocols test whether the findings repeat in this setting; external validity is outside their scope. Hardware, software versions and the course of each run are given in Sections S6, S7 and S9.

5. Results

5.1. Screening against padding

Table 3 gives the paired differences from conservative speed on the 46 development instances where conservative speed found a plan. The complete original procedure was 8.3–21.4 points more reliable, depending on size, although the interval at 200 customers includes zero. The slack-aware objective on its own, without screening, was less reliable than padding at every size, so the gain needs both the family of buffered candidates and the screen. Applying the same screen to conservative-speed candidates also helped, which makes a fixed 20% pad the weaker of the two padding baselines.

The advantage grows with noise when delays are independent (Fig. 1). For fixed plans with $\rho = 0$, the difference between the original procedure and conservative speed went from -3.1 points at $\sigma = 0.10$, where padding was slightly ahead, to 30.5 points at $\sigma = 0.30$. With strongly correlated delays the increase was not monotonic. Depending on size, the original procedure drove 2.0–4.9% more distance

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Table 3
Paired differences in service probability (percentage points) on the same subsets as Table S3 of the SI. SC denotes screened conservative speed with its prespecified fallback.

n	S minus C	P minus C	SC minus C
20	−11.0 [−16.1, −6.0]	21.4 [9.6, 31.3]	28.2 [21.4, 34.3]
50	−13.1 [−17.4, −8.9]	17.9 [2.8, 32.0]	11.8 [2.7, 22.2]
100	−16.2 [−20.6, −11.7]	16.0 [4.8, 24.8]	11.2 [2.8, 19.4]
200	−12.9 [−18.5, −5.7]	8.3 [−5.1, 20.0]	5.7 [0.0, 12.9]

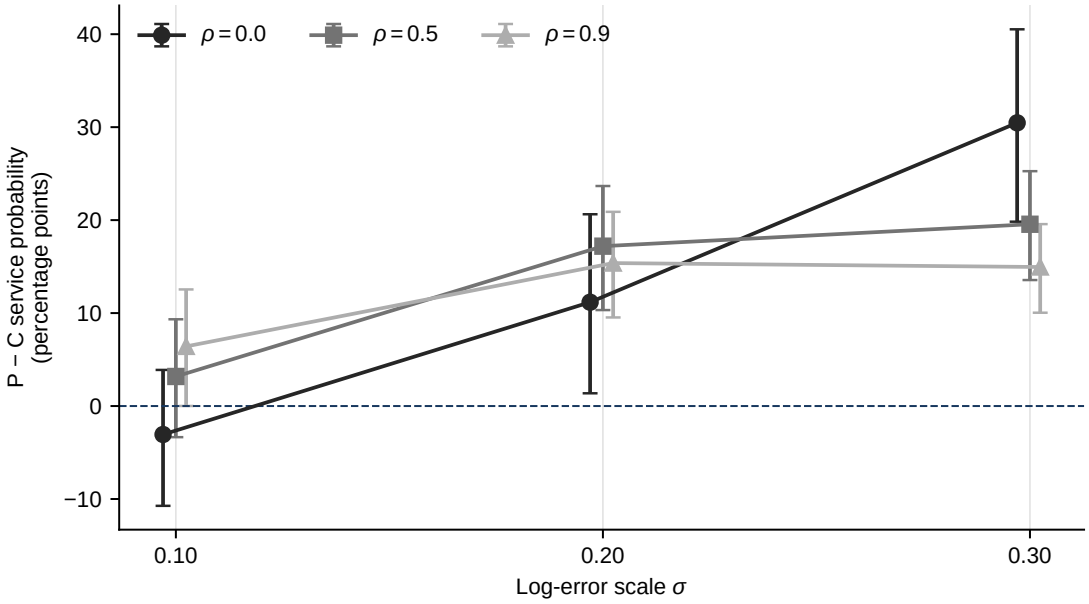


Fig. 1. Paired difference in service probability between the original procedure (P) and conservative speed (C) on the 46 common development instances, for fixed plans across the disturbance grid. Circles, squares and triangles mark $\rho = 0, 0.5$ and 0.9; error bars are 95% bootstrap intervals over instances, not simultaneous across cells.

than conservative speed, and its fleet was sometimes larger and sometimes smaller (Table S5 of the SI).

5.2. Capping the contraction

On 21 development instances the original procedure admitted no plan, and on 17 of them a reachability obstruction was certified at some buffer level. Capping (Table 4) raised the number of admitted instances from 32 to 42 and pooled service by 7.7 [3.5, 12.5] points. At 100 and 200 customers the capped procedure admitted a plan on 18/18 instances and pooled 97.5–97.6%. On average it used 2.5% more distance and 0.43 more vehicles than the original procedure, and 1.88 more vehicles at 200 customers.

Pooled service stays below the target because of the fallback. Plans admitted by the capped procedure

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Table 4
Development study, all 53 instances. For each procedure: instances admitted by screening out of all instances of that size, with the number whose lower bound also reached 0.95 on the validation bank in parentheses, and pooled validated service probability (%) with a 95% bootstrap interval, fallback plans included.

	Original procedure		Screened conservative		Capped procedure	
n	Admitted	Pooled	Admitted	Pooled	Admitted	Pooled
20	14/20 (14)	81.6 [70.2, 91.9]	15/20 (15)	87.7 [79.0, 94.7]	14/20 (14)	81.6 [70.3, 92.0]
50	8/15 (8)	72.7 [59.2, 86.4]	4/15 (4)	67.5 [58.2, 77.3]	10/15 (10)	80.1 [66.9, 90.8]
100	6/10 (5)	81.3 [68.8, 93.3]	4/10 (4)	77.5 [67.2, 88.2]	10/10 (9)	97.5 [96.7, 98.2]
200	4/8 (3)	80.6 [68.7, 92.3]	2/8 (2)	78.3 [71.0, 86.5]	8/8 (7)	97.6 [96.9, 98.2]

averaged 97.4% validated service, while its 11 fallback plans, all at 20 or 50 customers, averaged 45.2%. Of the 42 admitted plans, 40 also had a lower bound of at least 0.95 on the validation bank.

5.3. Confirmatory evidence

Table 5 lists every prespecified hypothesis with its estimate and decision. In protocol 1 the original procedure beat conservative speed by 15.7 points, and capping added another 11.7. Capping admitted 46 instances against 32 and lost none. H2 and H3 are not independent, because 11.6 of the 11.7 points in H2 come from instances that only the capped procedure admitted. Over all instances the capped procedure pooled 91.0%, against 79.4% for the original procedure.

Protocol 2 asked whether the gain holds up with a stronger generator. With HGS, screening with capped contraction beat HGS conservative-speed planning by 28.0 points, and pooled service was 91.6% against 58.4%. The HGS and OR-Tools capped procedures differed by -0.2 points $[-4.0, 3.7]$. No equivalence margin was set in advance, so this interval limits the size of the difference without establishing equivalence. Capping mattered more with HGS. At 100 and 200 customers it raised the number of admitted instances from 3 to 18 of 18. The conservative fallback, which protocol 1 had suggested after the fact, did not measurably improve the capped OR-Tools procedure (G2). That procedure fell back on only 8 instances, the new rule changed the plan on 4 of them, and the changes went both ways. The rule did help the uncapped procedure (G3).

5.4. Comparison with robust optimization

On the 45 development instances with up to 100 customers, the standard robust model under box uncertainty with $\theta = 0.6$ had no plan on 16 instances. In each case some customer could not be protected even on a direct trip, and every smaller budget failed on the same instances. With relief every setting had a plan, and moving between settings traded distance for service along a smooth curve (Fig. 2). Screened robust with relief pooled 97.2%, compared with 84.6% for the OR-Tools capped procedure and 93.6% for the HGS one, and it used 0.53 more vehicles than the HGS procedure.

Part A of protocol 3 confirmed these findings on the earlier confirmatory instances with up to 100 customers. Screened robust with relief beat the OR-Tools capped procedure by 6.7 points over all 90 of

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Table 5
Every prespecified hypothesis of the three protocols. Primary hypotheses are marked *. Differences are in percentage points of validated service-event probability unless stated; intervals are paired bootstrap intervals over instances (20,000 resamples), at 97.5% for the two primary hypotheses of each study (Bonferroni) and 95% otherwise. A hypothesis is confirmed when it meets the decision rule fixed in its protocol.

Hyp.	Contrast	Estimate [interval]	Level	Decision
Protocol 1: 53 new instances, OR-Tools generator				
H1*	Original procedure minus conservative speed (50 instances)	15.7 [7.5, 23.6]	97.5%	confirmed
H2*	Capped minus original procedure (53 instances)	11.7 [5.9, 17.9]	97.5%	confirmed
H3	Instances admitted, capped versus original procedure	46 vs 32 ($p = 0.00012$)	exact	confirmed
H4	Advantage over conservative speed at $\sigma = 0.30$ minus at $\sigma = 0.10$, $\rho = 0$	28.3 [19.6, 36.4]	95%	confirmed
Protocol 2: 53 new instances, HGS and OR-Tools generators				
G1*	HGS capped procedure minus HGS conservative speed (49 instances)	28.0 [23.0, 32.8]	97.5%	confirmed
G2*	Conservative minus slack-aware fallback, OR-Tools capped procedure	0.4 [−0.7, 2.1]	97.5%	not confirmed
G3	The same for the uncapped OR-Tools procedure	3.8 [1.6, 6.2]	95%	confirmed
G4	HGS minus OR-Tools capped procedure, both with conservative fallback	−0.2 [−4.0, 3.7]	95%	estimate
Protocol 3, part A: 90 confirmatory instances with up to 100 customers				
R1*	Screened robust with relief minus OR-Tools capped procedure	6.7 [3.0, 11.1]	97.5%	confirmed
R2*	Screened robust with relief minus HGS capped procedure (45 instances)	6.7 [2.2, 12.2]	97.5%	confirmed
R3	The same contrast in vehicles	0.42 [0.20, 0.67]	95%	confirmed
R4	Distance (%) of screened robust versus the most protective relieved setting	−1.3 [−1.7, −0.9]	95%	confirmed
Protocol 3, part B: 53 new instances, best-screened fallback				
F1*	Best-screened minus slack-aware fallback, OR-Tools capped procedure	7.3 [2.3, 13.2]	97.5%	confirmed
F2*	Best-screened minus conservative fallback, HGS capped procedure	4.2 [1.2, 8.0]	97.5%	confirmed
F3	Pooled service (%), OR-Tools capped procedure with the new rule	94.4 [90.1, 97.1]	95%	estimate
F4	Pooled service (%), HGS capped procedure with the new rule	94.9 [91.0, 97.1]	95%	estimate

them, and the HGS capped procedure by 6.7 points over the 45 instances of protocol 2, the only ones with HGS plans. The two estimates happen to coincide. Screened robust with relief pooled 97.2% and used 0.42 more vehicles than the HGS procedure. Within the robust family, screening returned plans 1.3% shorter than the most protective setting, at a cost of 1.2 points of service. When the robust plans were limited after the fact to no more vehicles than the comparator used, they were no longer measurably more reliable. The differences were 3.2 [−1.1, 8.1] points against the HGS capped procedure and −0.7 [−3.9, 2.7] against the OR-Tools one. On these instances the screened procedures used fewer vehicles at

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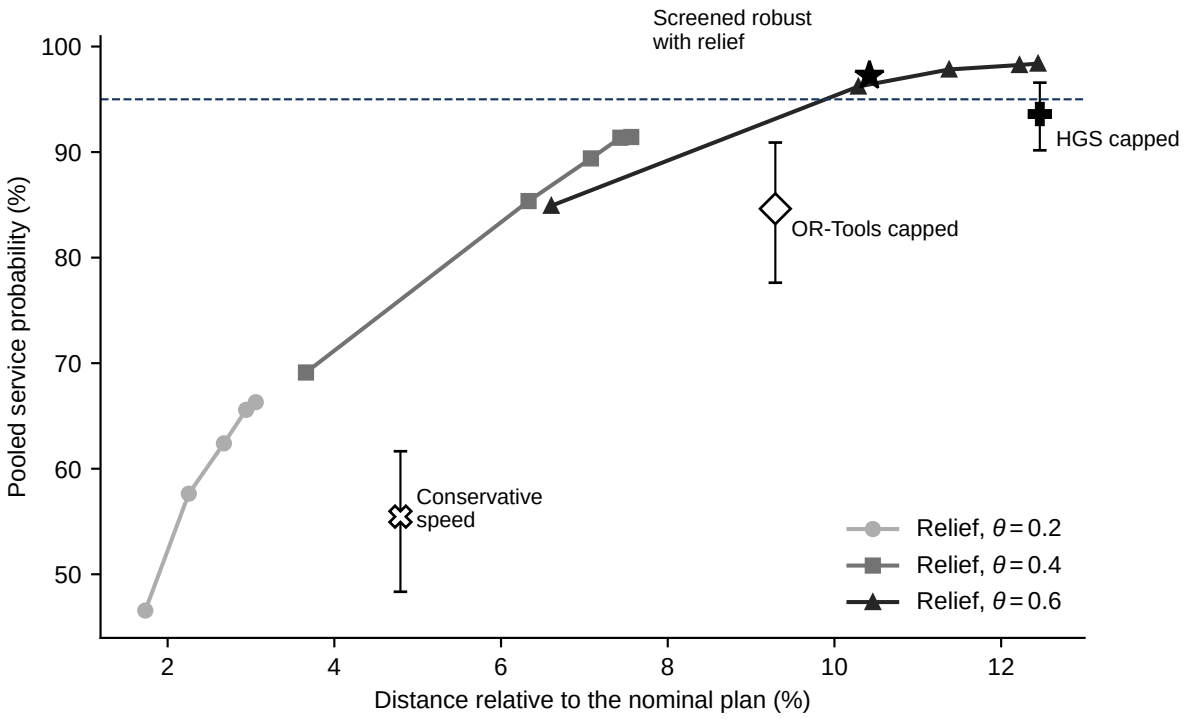


Fig. 2. Pooled validated service probability against mean distance relative to the nominal robust plan, on the 45 development instances with up to 100 customers. Lines join the relieved robust settings for each θ, from $\Gamma = 1$ to box uncertainty; the screened procedures and conservative speed are shown with 95% bootstrap intervals. The dashed line marks the target.

some cost in reliability. Results with 200 customers are in Section S10.12.

5.5. Fallback rule

After the first two protocols, the stored records showed that returning the best-screened candidate when nothing is admitted would have raised pooled service in every study and for every procedure, by 3.0 to 15.6 points. Part B of protocol 3 tested this rule on 53 new instances. Compared with each procedure's default fallback, it improved the OR-Tools capped procedure by 7.3 points and the HGS capped procedure by 4.2, bringing pooled service to 94.4% and 94.9%. Both estimates are just below the 95% target. On 1 of the 53 instances the anchor solves found no plan, so neither procedure could run and that instance scored zero. Without it, pooled service was 96.3% and 96.7% (descriptive). Plans returned by a fallback have not passed the screen.

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5.6. Sensitivity and further checks

Objective weights. With equal weights, fewer development instances were admitted (23 against 31 in the same run), and the advantage over conservative speed fell to 6.3 points [−1.1, 13.7]. A later run that repeated both weight vectors found the advantage falling from 17.5 to 8.1 points [1.1, 15.1]. Raising only the driver-duration weight changed service by −8.9 points, and raising only the fleet weight changed it by −5.6, and the run cannot separate the two effects. The stored candidates suggest that the advantage depends on how much waiting time the objective leaves in the routes to absorb delays. None of the protocols varied the weights (Section S10.8).

Benchmark instances. On 8 Solomon R1 and RC1 instances, OR-Tools found a plan for 5 and screening admitted 3. On 20 Gehring–Homberger instances with 200 customers (Gehring and Homberger, 1999), with HGS as the generator, the procedure could run on 18, admitted a plan on 17 and beat conservative speed by 15.1 points [11.0, 19.1] (Sections S10.9 and S10.11).

Delay-model misspecification. Under four alternative delay models, the fixed plans of the capped procedures lost up to 6.9 points, and they stayed ahead of conservative speed. When the stored candidates were screened again under each model and the best-screened fallback was used, pooled service stayed between 93.2 and 96.7% (Section S14.3).

Family-wise screening and computation time. Across all studies and procedures, a Bonferroni bound over the 13 buffer levels removed all 11 of the 318 admitted plans that later had a validation lower bound below 0.95, while 1 of the newly selected plans did. The cost was at most 2 admissions per study and procedure, and pooled service changed by between −1.8 and 0.2 points. Screening a candidate took 3 to 24 ms, whereas generating one took seconds (Sections S14.1 and S14.2).

6. Discussion and limitations

For a planner who already runs a VRPTW solver, the change is cheap. Generating a handful of buffered variants and simulating them costs little beyond the solves themselves, and it needs no change to the solver or its objective. On the same instances it gave plans that were measurably more reliable than a fixed 20% pad. Every admitted plan also comes with a tested lower bound on a defined service event, and the validation bank shows how the plan did on days it had not seen. The reachability cap is a one-line change that removed most certified failures, and it should be applied to any method that adds buffers, robust models included.

The evidence is limited in several ways. The instances are synthetic, and the disturbance parameters are assumed and have not been estimated from vehicle GPS data, so the numbers should not be read as forecasts for a real fleet. The protocols test repeatability with one instance generator and one delay model, and the Solomon and Gehring–Homberger checks are small. The OR-Tools generator is weaker than the state of the art, which is why protocol 2 repeated the test with HGS. Even so, two generators cannot show that the result holds for every solver. The robust comparison covers one budgeted model with one search configuration, and distributionally robust and riskiness-index methods (Jaillet et al., 2016; Zhang et al., 2019) were not tested. The advantage depends on the objective weights, and equal weights more than halved it. The Wilson bound holds for each candidate separately and gives no simultaneous guarantee over the grid. Finally, keeping every departure fixed makes the evaluation harsher than an operator who

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adjusts departures would face. Breakdowns, new orders and rolling-horizon replanning are left out, so the study says nothing about how to recover once a delay has happened.

Under the prespecified fallback rules, pooled service of the OR-Tools and HGS procedures stayed below 95% in every study, because some instances had no admitted plan. The best-screened fallback narrows that gap, but closing it needs better candidates on those instances, where a stronger or a stochastic generator could help most.

7. Conclusion

On matched synthetic instances, screening a family of buffered plans from a standard VRPTW solver gave more reliable service than padding every trip, with OR-Tools and with HGS, and three time-stamped protocols supported the main findings. One primary hypothesis was not confirmed. Returning the conservative-speed plan when nothing is admitted did not measurably help the capped OR-Tools procedure, although returning the best-screened candidate did. Capping each customer's buffer at its direct-reachability limit removed a failure that also prevents robust models from finding any plan. A budgeted-robust optimizer with the same relief was more reliable still, but it used more vehicles, and at the comparator's fleet size its advantage was not statistically detectable. The screening layer and the reachability relief worked with every planner tested. The study has yet to be repeated on a real road network with delay parameters estimated from fleet data.

Data availability

All code, generated instances, raw records and analysis scripts are openly available on Zenodo (Ali, 2026d); code is released under the MIT License and records under CC BY 4.0. The three protocol freezes are separate Zenodo records (Ali, 2026a,b,c). The Solomon and Gehring–Homberger instance files are third-party data (Solomon, 1987; ML4VRP, 2024; Gehring and Homberger, 1999) and remain under their original terms.

Use of generative AI

Generative AI assistants (OpenAI's ChatGPT and Anthropic's Claude) were used for code development, code review, checking the analysis, and drafting and editing text. They are not authors. Every experiment was run from the archived code and every reported number regenerates from the archived records. The author reviewed all code and text and takes full responsibility for the content.

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